A.V.KAVERI,

192512188.

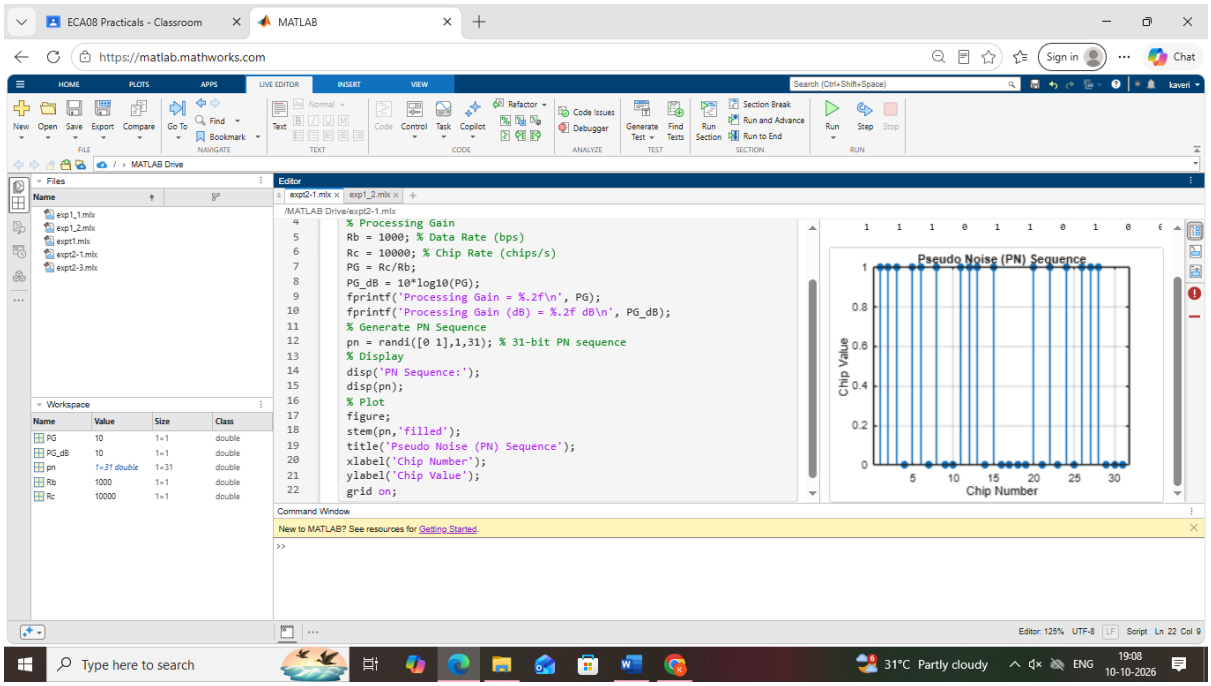
EXPERIMENT – 27

CALCULATION OF PROCESSING GAIN AND GENERATION OF PN SEQUENCES FOR SPREAD SPECTRUM COMMUNICATION USING MATLAB SIMULATION

PROGRAM:

```
clc;
clear;
close all;
% Processing Gain
Rb = 1000; % Data Rate (bps)
Rc = 10000; % Chip Rate (chips/s)
PG = Rc/Rb;
PG_dB = 10*log10(PG);
fprintf('Processing Gain = %.2f\n', PG);
fprintf('Processing Gain (dB) = %.2f dB\n', PG_dB);
% Generate PN Sequence
pn = randi([0 1],1,31); % 31-bit PN sequence
% Display
disp('PN Sequence:');
disp(pn);
% Plot
figure;
stem(pn,'filled');
title('Pseudo Noise (PN) Sequence');
xlabel('Chip Number');
ylabel('Chip Value');
grid on;
```

OUTPUT:


ECA08 Practicals - Classroom
MATLAB
https://matlab.mathworks.com
Sign in
Chat
HOME
PLOTS
APPS
LIVE EDITOR
INSERT
VIEW
MATLAB Drive
Files
Name
exp1_1.mlx
exp1_2.mlx
expt1.mlx
expt2-1.mlx
expt2-3.mlx
Workspace
Name
Value
Size
Class
PG
10
1×1
double
PG_dB
10
1×1
double
pn
1×31 double
1×31
double
Rb
1000
1×1
double
Rc
10000
1×1
double
Editor
expt2-1.mlx
exp1_2.mlx
/MATLAB Drive/expt2-1.mlx
% Processing Gain
Rb = 1000; % Data Rate (bps)
Rc = 10000; % Chip Rate (chips/s)
PG = Rc/Rb;
PG_dB = 10*log10(PG);
fprintf('Processing Gain = %.2f\n', PG);
fprintf('Processing Gain (dB) = %.2f dB\n', PG_dB);
% Generate PN Sequence
pn = randi([0 1],1,31); % 31-bit PN sequence
% Display
disp('PN Sequence:');
disp(pn);
% Plot
figure;
stem(pn,'filled');
title('Pseudo Noise (PN) Sequence');
xlabel('Chip Number');
ylabel('Chip Value');
grid on;
Pseudo Noise (PN) Sequence
Chip Value
Chip Number
Command Window
New to MATLAB? See resources for Getting Started.
Type here to search
31°C Partly cloudy
ENG
19:08
10-10-2026